Summary of Backpropagation-Free Training

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1 Review of Experimental Design

In last week's meeting, I presented the experimental results of training models using a hybrid approach where the client and server sides were trained with BP-free and BP methods, respectively. Unfortunately, as shown in Table 1, the accuracy achieved by this method was significantly lower than expected. Moreover, this approach failed to reduce either the training time or the amount of information transmitted during the process.

Table 1: Accuracy of split learning for different training methods.

Training Method	**Model**	**Accuracy (%)**
All layers use BP	ResNet-5	70.94
	VGG-8	76.84
All layers use BP-Free	ResNet-5	65.07
	VGG-8	76.08
BP-Free=1	ResNet-5	25.51
	VGG-8	33.46
BP-Free=2	ResNet-5	23.17
	VGG-8	29.66

Given these unsatisfactory results, we abandoned the mixed use of BP and BP-free methods. Instead, we opted to apply the BP-free method exclusively on both the client and server sides. Additionally, we modified the BP-free method by introducing noise only at the server side rather than adding noise to the output of each layer. This adjustment is expected to reduce client-side training time and communication overhead.

2 Experimental Results

The experimental setup remains consistent with the previous one. The models used are ResNet-5 and VGG-8, with either the first or the first two layers placed on the client side. The experimental results are summarized in Tables 2 and 3. Table 2 presents the accuracy results under different configurations, while Table 3 shows the corresponding communication loads.

Table 2: Accuracy of split learning for different cutting positions.

Client-side Layers (L_c)	**Model**	**Accuracy (%)**
$L_c = 1$	ResNet-5	58.48
	VGG-8	73.17
$L_c = 2$	ResNet-5	49.13
	VGG-8	69.60

From the results presented in Tables 2 and 3, it can be observed that as L_c increases, both the model accuracy and communication load are affected. Specifically, the accuracy decreases because fewer layers are subject to perturbations, while the communication load increases due to the larger size of the layers at the cutting point. Therefore, optimal performance is achieved when $L_c = 1$.

Table 3: Communication load for different cutting positions.

Client-side Layers (L_c)	**Model**	**Communication Load (KB)**
$L_c = 1$	ResNet-5	4098
	VGG-8	8194
$L_c = 2$	ResNet-5	8194
	VGG-8	8194

Under the same layering configuration, the communication load of our proposed method is reduced by 50% and 99%, respectively, compared to the original BP and BP-free approaches. The original BP method requires transmitting both activation values and gradients during the forward and backward, resulting in high communication overhead. In contrast, the original BP-free method avoids gradient computation but involves repeatedly transmitting the same information over 100 times, introducing a new communication burden.

References